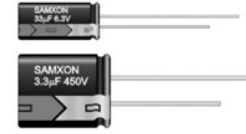


## FEATURES

1. Rated working voltage range 6.3 to 100V DC/160 to 450V DC at operation temperature range -40 to +105°C/-25 to +105°C.
2. This series is for communication equipments, switching power supply, industrial measuring instruments, automotive electric products, etc.



## SPECIFICATIONS

| Item                                                        | Performance Characteristics                                                                                                                   |                                                           |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Operating Temperature Range                                 | -40 to +105°C                                                                                                                                 | -25 to +105°C                                             |
| Rated Working Voltage Range                                 | 6.3 to 100V                                                                                                                                   | 160 to 450V                                               |
| Nominal Capacitance Range                                   | 0.1 to 33000µF                                                                                                                                |                                                           |
| Capacitance Tolerance                                       | ±20% (120Hz, +20°C)                                                                                                                           |                                                           |
| Leakage Current                                             | $I \leq 0.01CV$ or $3(\mu A)$ whichever is greater measured after 2 minutes application of rated working voltage at +20°C                     |                                                           |
| tan δ (120Hz, +20°C)                                        | $I \leq 0.03CV + 40(\mu A)$ after 2 minutes application of rated working voltage at +20°C                                                     |                                                           |
| tan δ (120Hz, +20°C)                                        | Working Voltage (V)                                                                                                                           | 6.3    10    16    25    35    50    63    100            |
|                                                             | tan δ (max.)                                                                                                                                  | 0.26   0.22   0.18   0.16   0.14   0.12   0.10   0.08     |
|                                                             | Working Voltage (V)                                                                                                                           | 160    200    220    250    350    400    420    450      |
|                                                             | tan δ (max.)                                                                                                                                  | 0.20   0.20   0.20   0.20   0.24   0.24   0.24   0.24     |
| For capacitance value > 1000µF, add 0.02 per another 1000µF |                                                                                                                                               |                                                           |
| Low Temperature Characteristics                             | Impedance ratio max. at 120Hz                                                                                                                 |                                                           |
|                                                             | Working Voltage (V)                                                                                                                           | 6.3    10    16    25    35    50    63    100            |
|                                                             | Z-25°C / Z+20°C                                                                                                                               | 5    4    3    2    2    2    2    2                      |
|                                                             | Z-40°C / Z+20°C                                                                                                                               | 10    8    6    4    3    3    3    3                     |
| Low Temperature Characteristics                             | Working Voltage (V)                                                                                                                           | 160    200    220    250    350    400    420    450      |
|                                                             | Z-25°C / Z+20°C                                                                                                                               | 3    3    3    4    4    6    6    15                     |
|                                                             | For capacitance value > 1000µF, add 0.5 per another 1000µF for Z-25°C / Z+20°C                                                                |                                                           |
|                                                             | add 1.0 per another 1000µF for Z-40°C / Z+20°C                                                                                                |                                                           |
| High Temperature Loading                                    | Test conditions                                                                                                                               | Post test requirements at +20°C                           |
|                                                             | Duration : $\phi D \leq 6.3 \geq 8$                                                                                                           | Leakage current : $\leq$ Initial specified value          |
|                                                             | Load life 1000h 2000h                                                                                                                         | Cap. change : within $\pm 20\%$ of initial measured value |
|                                                             | Ambient temp. : +105°C                                                                                                                        | tan δ : $\leq 200\%$ of initial specified value           |
| High Temperature Loading                                    | Applied voltage : DC voltage with maximum permissible ripple current specified at +105°C                                                      |                                                           |
|                                                             | (Sum of the DC voltage and super-imposed peak AC voltage for maximum permissible ripple current should be equal to rated DC working voltage). |                                                           |
|                                                             |                                                                                                                                               |                                                           |
|                                                             |                                                                                                                                               |                                                           |
| Shelf Life                                                  | Test conditions                                                                                                                               | Post test requirements at +20°C                           |
|                                                             | Duration : 1000 hours                                                                                                                         | Leakage current : $\leq$ Initial specified value          |
|                                                             | Ambient temp. : +105°C                                                                                                                        | Cap. change : within $\pm 20\%$ of initial measured value |
|                                                             | Applied voltage : (None)                                                                                                                      | tan δ : $\leq 200\%$ of initial specified value           |
| Others                                                      | JIS C - 5101 (IEC 60384)                                                                                                                      |                                                           |

## CASE SIZE TABLE

Safety vent for  $\phi \geq 6.3$

Technical drawing of a radial bulk capacitor. The side view shows a cylindrical body with a safety vent on top. Dimensions include: total length  $L + \alpha \text{ max}$ , distance from body to vent  $15 \text{ min}$ , vent length  $4 \text{ min}$ , vent diameter  $\phi d \pm 0.05$ , and vent thickness  $F \pm 0.5$ . The end view shows the outer diameter  $\phi D + \beta \text{ max}$ . The unit is mm.

| $\phi D$ | 5            | 6.3 | 8   | 10  | 12.5              | 16  | 18  | 22   | 25   |  |  |  |
|----------|--------------|-----|-----|-----|-------------------|-----|-----|------|------|--|--|--|
| F        | 2.0          | 2.5 | 3.5 | 5.0 | 5.0               | 7.5 | 7.5 | 10.0 | 12.5 |  |  |  |
| $\phi d$ | 0.5          |     |     | 0.6 |                   | 0.8 |     |      | 1.0  |  |  |  |
| $\alpha$ | (L < 20) 1.5 |     |     |     | (L $\geq$ 20) 2.0 |     |     |      |      |  |  |  |
| $\beta$  | (D < 20) 0.5 |     |     |     | (D $\geq$ 20) 1.0 |     |     |      |      |  |  |  |

## PART NUMBER SYSTEM (EXAMPLE: 25V330 µF)

|   |   |   |     |   |   |   |   |    |    |    |    |    |    |
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| 1 | 2 | 3 | 4   | 5 | 6 | 7 | 8 | 9  | 10 | 11 | 12 | 13 | 14 |
| E | K | M | 337 | M | 1 | E | F | 12 | R  | R  |    |    |    |
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+105°C, General(普通品)

## STANDARD RATINGS

| Voltage (Code) |      | 6.3V (0J) |                | 10V (1A)  |                | 16V (1C)  |                | 25V (1E)  |                |
|----------------|------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|
| Cap.(μF)       | Code | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current |
| 0.1            | 104  |           |                |           |                |           |                |           |                |
| 0.15           | 154  |           |                |           |                |           |                |           |                |
| 0.22           | 224  |           |                |           |                |           |                |           |                |
| 0.33           | 334  |           |                |           |                |           |                |           |                |
| 0.47           | 474  |           |                |           |                |           |                |           |                |
| 1              | 105  |           |                |           |                |           |                |           |                |
| 2.2            | 225  |           |                |           |                |           |                |           |                |
| 3.3            | 335  |           |                |           |                |           |                |           |                |
| 4.7            | 475  |           |                |           |                |           |                | 5 x 11    | 26             |
| 10             | 106  |           |                |           |                | 5 x 11    | 35             | 5 x 11    | 38             |
| 22             | 226  |           |                | 5 x 11    | 49             | 5 x 11    | 54             | 5 x 11    | 57             |
| 33             | 336  | 5 x 11    | 54             | 5 x 11    | 60             | 5 x 11    | 64             | 5 x 11    | 75             |
| 47             | 476  | 5 x 11    | 65             | 5 x 11    | 70             | 5 x 11    | 80             | 5 x 11    | 84             |
| 68             | 686  | 5 x 11    | 70             | 5 x 11    | 75             | 5 x 11    | 90             | 5 x 11    | 92             |
| 100            | 107  | 5 x 11    | 95             | 5 x 11    | 105            | 5 x 11    | 125            | 6.3 x 11  | 159            |
| 220            | 227  | 5 x 11    | 153            | 5 x 11    | 170            | 6.3 x 11  | 213            | 8 x 12    | 285            |
| 330            | 337  | 6.3 x 11  | 216            | 6.3 x 11  | 239            | 8 x 12    | 315            | 8 x 12    | 340            |
| 470            | 477  | 6.3 x 11  | 258            | 6.3 x 11  | 285            | 8 x 12    | 366            | 10 x 12.5 | 471            |
| 680            | 687  | 8 x 12    | 365            | 8 x 12    | 408            | 10 x 12.5 | 480            | 10 x 16   | 620            |
| 1000           | 108  | 8 x 12    | 443            | 10 x 12.5 | 571            | 10 x 16   | 680            | 10 x 20   | 821            |
| 2200           | 228  | 10 x 16   | 740            | 10 x 20   | 886            | 12.5 x 20 | 1108           | 12.5 x 20 | 1176           |
| 3300           | 338  | 10 x 20   | 1032           | 12.5 x 20 | 1205           | 12.5 x 25 | 1389           | 16 x 25   | 1646           |
| 4700           | 478  | 12.5 x 20 | 1280           | 12.5 x 25 | 1492           | 16 x 25   | 1740           | 16 x 30   | 2012           |
| 6800           | 688  | 12.5 x 25 | 1554           | 16 x 25   | 1824           | 16 x 30   | 2081           | 16 x 35   | 2308           |
| 10000          | 109  | 16 x 25   | 1897           | 16 x 30   | 1980           | 16 x 35   | 2379           | 18 x 35   | 2500           |
| 15000          | 159  | 16 x 30   | 2188           | 16 x 40   | 2180           | 18 x 35   | 2600           |           |                |
| 22000          | 229  | 18 x 35   | 2400           | 18 x 40   | 2407           |           |                |           |                |
| 33000          | 339  | 18 x 40   | 2555           |           |                |           |                |           |                |

Maximum Allowable Ripple Current (mA rms) at 105°C 120Hz

Case Size φD x L(mm)

| Voltage (Code) |      | 35V (1V)  |                | 50V (1H)  |                | 63V (1J)  |                | 100V (2A) |                |
|----------------|------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|
| Cap.(μF)       | Code | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current |
| 0.1            | 104  |           |                | 5 x 11    | 1              |           |                |           |                |
| 0.15           | 154  |           |                | 5 x 11    | 1.5            |           |                |           |                |
| 0.22           | 224  |           |                | 5 x 11    | 3              |           |                |           |                |
| 0.33           | 334  |           |                | 5 x 11    | 4              |           |                |           |                |
| 0.47           | 474  |           |                | 5 x 11    | 7              |           |                | 5 x 11    | 10             |
| 1              | 105  |           |                | 5 x 11    | 13             |           |                | 5 x 11    | 16             |
| 2.2            | 225  |           |                | 5 x 11    | 20             |           |                | 5 x 11    | 23             |
| 3.3            | 335  |           |                | 5 x 11    | 30             |           |                | 5 x 11    | 34             |
| 4.7            | 475  | 5 x 11    | 28             | 5 x 11    | 37             | 5 x 11    | 40             | 5 x 11    | 40             |
| 10             | 106  | 5 x 11    | 41             | 5 x 11    | 54             | 5 x 11    | 59             | 6.3 x 11  | 61             |
| 22             | 226  | 5 x 11    | 67             | 5 x 11    | 79             | 5 x 11    | 79             | 6.3 x 11  | 92             |
| 33             | 336  | 5 x 11    | 80             | 5 x 11    | 101            | 6.3 x 11  | 122            | 8 x 12    | 144            |
| 47             | 476  | 5 x 11    | 101            | 6.3 x 11  | 133            | 6.3 x 11  | 146            | 10 x 12.5 | 199            |
| 68             | 686  |           |                |           |                | 8 x 12    | 155            | 10 x 16   | 240            |
| 100            | 107  | 6.3 x 11  | 168            | 8 x 12    | 229            | 10 x 12.5 | 251            | 10 x 20   | 349            |
| 220            | 227  | 8 x 12    | 294            | 10 x 16   | 509            | 10 x 20   | 504            | 12.5 x 25 | 662            |
| 330            | 337  | 10 x 12.5 | 419            | 10 x 16   | 589            | 12.5 x 20 | 688            | 12.5 x 25 | 800            |
| 470            | 477  | 10 x 16   | 547            | 10 x 20   | 707            | 12.5 x 20 | 810            | 16 x 25   | 990            |
| 680            | 687  | 10 x 20   | 682            | 12.5 x 20 | 923            | 12.5 x 25 | 1160           | 16 x 30   | 1289           |
| 1000           | 108  | 12.5 x 20 | 1023           | 12.5 x 25 | 1287           | 16 x 25   | 1448           | 18 x 40   | 2020           |
| 2200           | 228  | 16 x 25   | 1497           | 16 x 35   | 1884           | 18 x 35   | 1781           |           |                |
| 3300           | 338  | 16 x 30   | 1808           | 18 x 35   | 2167           |           |                |           |                |
| 4700           | 478  | 18 x 35   | 2335           |           |                |           |                |           |                |
| 6800           | 688  | 18 x 40   | 2400           |           |                |           |                |           |                |

Maximum Allowable Ripple Current (mA rms) at 105°C 120Hz

Case Size φD x L(mm)

Specifications are subject to change without notice. Should a safety or technical concern arise regarding the product, please be sure to contact our sales offices or agents immediately.

+105°C, General(普通品)

## STANDARD RATINGS

| Voltage (Code) |      | 160V (2C) |                | 200V (2D) |                | 220V (2N) |                | 250V (2E) |                |
|----------------|------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|
| Cap.(μF)       | Code | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current |
| 0.47           | 474  |           |                |           |                |           |                | 6.3 x 11  | 8              |
| 1              | 105  |           |                |           |                |           |                | 6.3 x 11  | 17             |
| 2.2            | 225  |           |                |           |                |           |                | 6.3 x 11  | 27             |
| 3.3            | 335  |           |                | 6.3 x 11  | 30             | 6.3 x 11  | 30             | 6.3 x 11  | 35             |
| 4.7            | 475  | 6.3 x 11  | 41             | 6.3 x 11  | 40             | 8 x 12    | 40             | 8 x 12    | 45             |
| 10             | 106  | 8 x 12    | 60             | 10 x 12.5 | 72             | 10 x 12.5 | 70             | 10 x 12.5 | 75             |
| 22             | 226  | 10 x 16   | 110            | 10 x 16   | 113            | 10 x 20   | 125            | 10 x 20   | 130            |
| 33             | 336  | 10 x 20   | 156            | 10 x 20   | 165            | 12.5 x 20 | 165            | 12.5 x 20 | 184            |
| 47             | 476  | 10 x 20   | 195            | 10 x 20   | 194            | 12.5 x 20 | 220            | 12.5 x 25 | 238            |
| 68             | 686  | 12.5 x 20 | 250            | 12.5 x 25 | 250            | 12.5 x 25 | 245            | 16 x 20   | 246            |
| 82             | 826  | 12.5 x 25 | 310            | 10 x 30   | 320            | 12.5 x 30 | 280            | 16 x 25   | 351            |
| 100            | 107  | 12.5 x 25 | 360            | 16 x 25   | 386            | 16 x 25   | 335            | 16 x 25   | 390            |
| 150            | 157  | 12.5 x 30 | 380            | 16 x 25   | 525            | 16 x 30   | 365            | 18 x 30   | 440            |
| 180            | 187  | 12.5 x 35 | 420            | 12.5 x 35 | 560            | 16 x 35   | 500            | 18 x 35   | 469            |
| 220            | 227  | 16 x 30   | 680            | 16 x 30   | 643            | 16 x 40   | 615            | 18 x 35   | 485            |
| 270            | 277  | 16 x 30   | 728            | 18 x 30   | 740            |           |                |           |                |
| 330            | 337  | 18 x 35   | 830            | 18 x 30   | 808            |           |                |           |                |
| 390            | 397  | 18 x 35   | 850            | 18 x 35   | 904            |           |                |           |                |
| 470            | 477  | 18 x 40   | 880            | 18 x 40   | 1016           |           |                |           |                |
| 560            | 567  | 18 x 45   | 925            | 18 x 45   | 1112           |           |                |           |                |

Maximum Allowable Ripple Current (mA rms) at 105°C 120Hz

Case Size φD x L(mm)

| Voltage (Code) |      | 350V (2V) |                | 400V (2G) |                | 420V (2M) |                | 450V (2W) |                |
|----------------|------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|
| Cap.(μF)       | Code | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current | Case Size | Ripple Current |
| 0.47           | 474  | 6.3 x 11  | 8              |           |                |           |                |           |                |
| 1              | 105  | 6.3 x 11  | 18             | 6.3 x 11  | 19             | 6.3 x 11  | 15             | 6.3 x 11  | 16             |
| 2.2            | 225  | 6.3 x 11  | 25             | 8 x 12    | 30             | 8 x 12    | 29             | 8 x 12    | 24             |
| 3.3            | 335  | 8 x 12    | 40             | 8 x 12    | 35             | 8 x 12    | 35             | 8 x 12    | 29             |
| 4.7            | 475  | 8 x 12    | 43             | 8 x 12    | 40             | 10 x 16   | 52             | 10 x 16   | 42             |
| 10             | 106  | 10 x 16   | 73             | 10 x 16   | 78             | 10 x 20   | 85             | 12.5 x 20 | 84             |
| 18             | 186  | 12.5 x 20 | 100            | 12.5 x 20 | 105            | 12.5 x 25 | 124            | 10 x 30   | 108            |
| 22             | 226  | 12.5 x 20 | 150            | 12.5 x 20 | 148            | 12.5 x 25 | 140            | 12.5 x 25 | 131            |
| 27             | 276  | 12.5 x 25 | 177            | 10 x 30   | 192            | 12.5 x 25 | 170            | 12.5 x 30 | 164            |
| 33             | 336  | 16 x 25   | 200            | 12.5 x 25 | 193            | 16 x 25   | 200            | 16 x 25   | 237            |
| 39             | 396  | 16 x 25   | 258            | 16 x 25   | 251            | 12.5 x 30 | 248            | 12.5 x 35 | 256            |
| 47             | 476  | 16 x 25   | 265            | 12.5 x 30 | 266            | 12.5 x 35 | 288            | 16 x 30   | 305            |
| 56             | 566  | 16 x 30   | 280            | 12.5 x 35 | 336            | 12.5 x 40 | 344            | 16 x 30   | 352            |
| 68             | 686  | 16 x 30   | 288            | 16 x 30   | 396            | 16 x 30   | 408            | 18 x 30   | 366            |
| 82             | 826  | 18 x 30   | 372            | 18 x 30   | 443            | 16 x 35   | 456            | 18 x 30   | 440            |
| 100            | 107  | 18 x 35   | 460            | 18 x 30   | 489            | 18 x 35   | 488            | 18 x 35   | 490            |
| 120            | 127  |           |                | 18 x 35   | 570            | 18 x 40   | 528            | 18 x 40   | 592            |
| 150            | 157  |           |                | 18 x 40   | 616            | 18 x 45   | 568            | 18 x 45   | 640            |
| 180            | 187  |           |                | 18 x 50   | 704            |           |                |           |                |

Maximum Allowable Ripple Current (mA rms) at 105°C 120Hz

Case Size φD x L(mm)

## RIPPLE CURRENT MULTIPLIER

| Frequency Coefficient |            |             |      |      |      |      |
|-----------------------|------------|-------------|------|------|------|------|
| Rated Voltage(V)      | Cap(μF)    | Coefficient | 50   | 120  | 300  | 1k   |
| 6.3 ~ 100             | ~ 47       | 0.75        | 1.00 | 1.35 | 1.57 | 2.00 |
|                       | 68 ~ 470   | 0.80        | 1.00 | 1.23 | 1.34 | 1.50 |
|                       | ≥ 560      | 0.85        | 1.00 | 1.10 | 1.13 | 1.15 |
| 160 ~ 450             | 0.47 ~ 220 | 0.80        | 1.00 | 1.25 | 1.40 | 1.60 |
|                       | ≥ 270      | 0.90        | 1.00 | 1.10 | 1.13 | 1.15 |

Specifications are subject to change without notice. Should a safety or technical concern arise regarding the product, please be sure to contact our sales offices or agents immediately.

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Miniature Aluminum Electrolytic Capacitors